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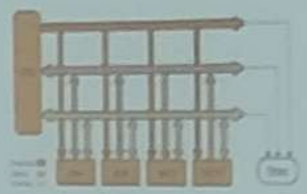
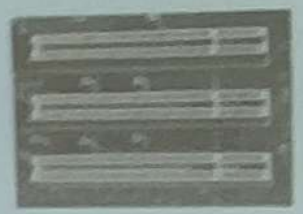
Internal CPU Interconnection

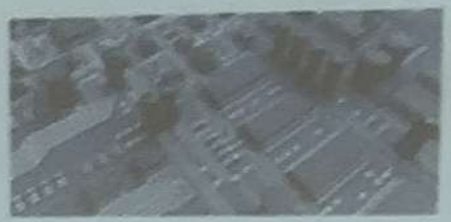
- A modern computer system is an interconnection of one or more processors, memory units, and input/output (I/O) devices.
- CPU interconnection is a mechanism that provides for communication among the control unit, ALU and registers.
- Alternatively, System interconnection is a mechanism that provides for communication among CPU, main memory and I/O.
- Another form of CPU interconnection is a **Bus**.
- A CPU interconnect system permits swapping of different CPUs in a computer without requiring that the different CPUs each be adapted to conform to a standard input/output bus.
- A personal computer (i.e., a microcomputer) additionally may include an optional special-purpose or custom processor used as an accelerator (e.g., GPU, FPGA).
- Translation between CPU input/output lines and a system bus is accomplished by a programmable logic circuit.



BUS

- *A bus is a collection of electrical wires that transfers data between components inside a computer.*
- *A bus is a communication system that transfers data between components inside a computer, or between computers.*
- The CPU communicates with memory and devices by a bus.
- A bus is a common pathway through which information flows from one component to another.
- One of the major roles of the bus is to provide an interface to memory.





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Functions of Bus

1. **Data sharing**
   - All types of buses found on a computer must be able to transfer data between the computer peripherals connected to it.
   - The data is transferred in in either serial or parallel, which allows the exchange of 1, 2, 4 or even 8 bytes of data at a time.
   - A byte is a group of 8 bits.
   - Buses are classified depending on how many bits they can move at the same time, which means that we have 8-bit, 16-bit, 32-bit or even 64-bit buses.
2. **Addressing**
   - A bus has address lines, which match those of the processor.
   - This allows data to be sent to or from specific memory locations.
3. **Power** - A bus supplies power to various peripherals that are connected to it.
4. **Timing** - The bus provides a system clock signal to synchronize the peripherals attached to it with the rest of the system.

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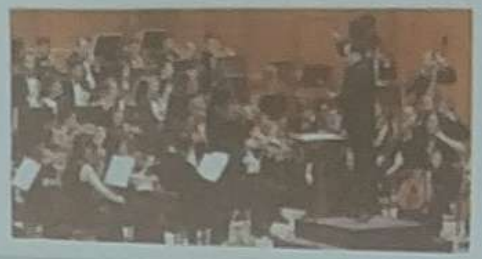
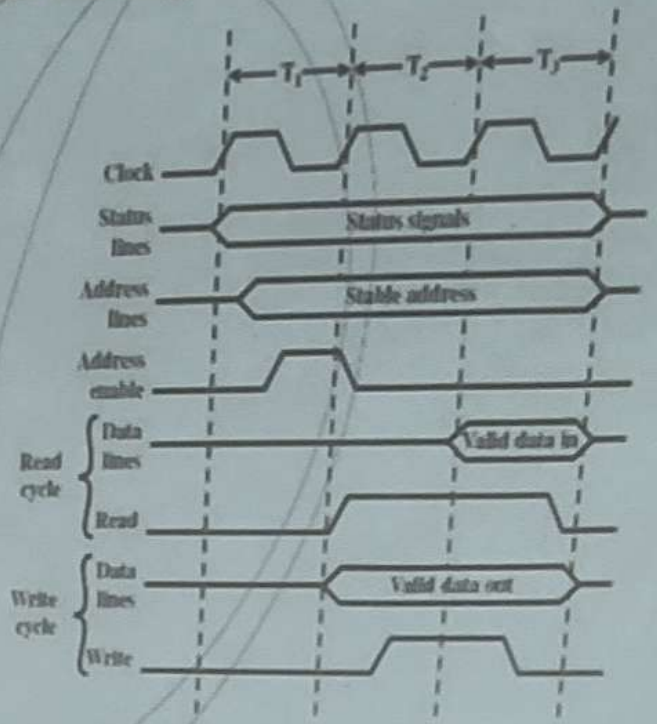
Bus Timing

- Co-ordination of events on bus.
- **Synchronous** – controlled by a clock.
- **Asynchronous** – timing is handled by well-defined specifications, i.e., a response is delivered within a specified time after a request.

Synchronous Bus Timing

- Events determined by clock signals
- Control Bus includes clock line
- A single 1-0 cycle is a bus cycle
- All devices can read clock line
- Usually sync on leading/rising edge
- Usually a single cycle for an event
- Analogy – *Orchestra conductor with baton*
- Usually stricter in terms of its timing requirements

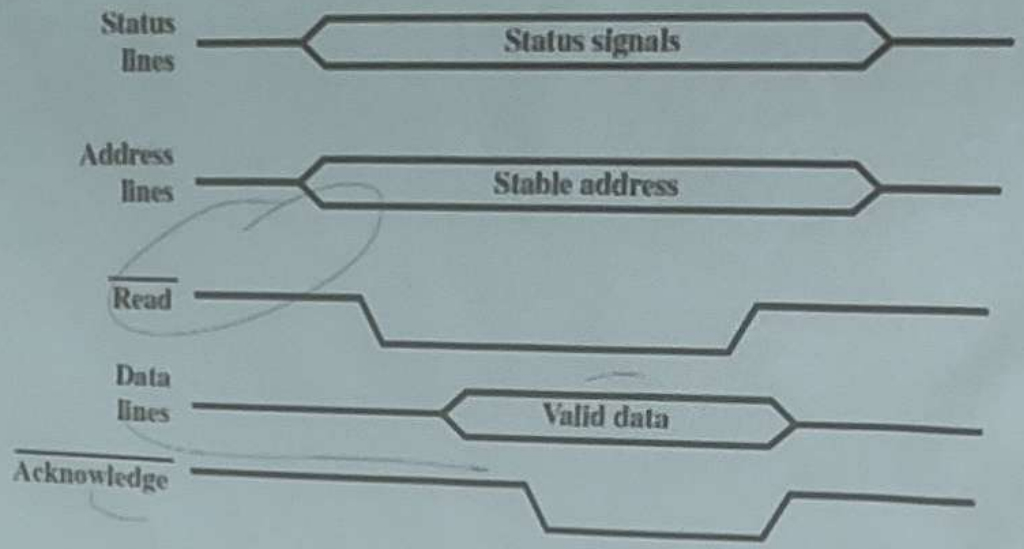




Asynchronous Timing

- Devices must have certain tolerances to provide responses to signal stimuli.
- More flexible allowing slower devices to communicate on same bus with faster devices.
- Performance of faster devices, however, is limited to speed of bus.
- Operations- Read and Write.

Asynchronous Timing – Read



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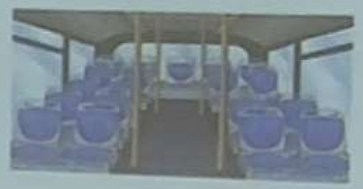
Bus Width

- *The size of a bus, known as its width.*
- It is important because it determines how much data can be transmitted at one time.
- For example, a 32-bit bus can transmit 32 bits of data, whereas a 64-bit bus can transmit 64 bits of data.
- *Every bus has a clock speed measured in MHz or GHz.*
- A fast bus allows data to be transferred faster, which makes applications run faster.
- Wider the bus the better the data transfer rate or the wider the addressable memory space
- Serial "width" is determined by length/duration of frame



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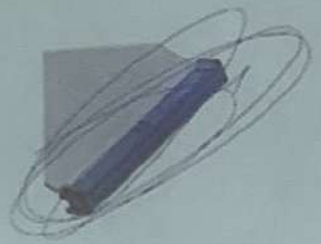
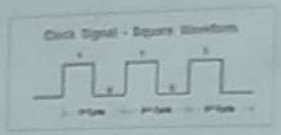
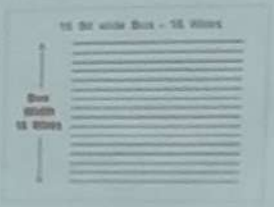
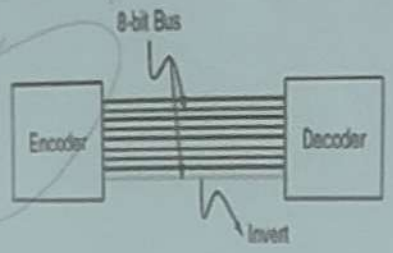
Bus Width









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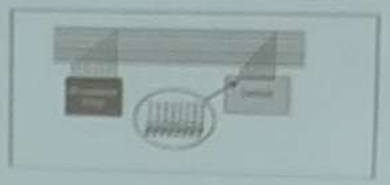
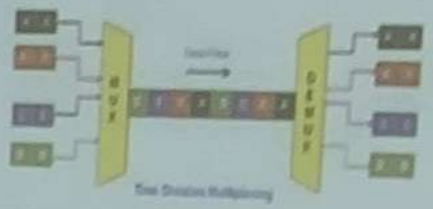
Types of Computer Buses

- A computer bus can be classified in three ways.
  - Dedicated vs. Time Multiplexed bus.
  - System bus
  - I/O bus

Bus: Dedicated vs. Time Multiplexed

- Dedicated
  - Separate data & address lines
- Time multiplexed
  - Shared lines
  - Address valid or data valid control line
  - Advantage - fewer lines
  - Disadvantages
    - More complex control
    - Degradation of performance





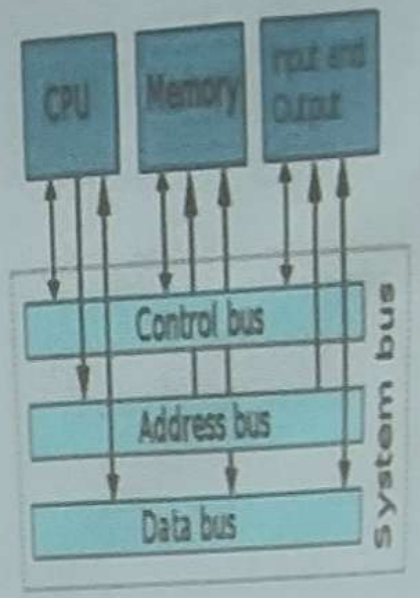
System Bus

- *The system bus is a pathway composed of cables and connectors used to carry data between a computer microprocessor and the main memory.*
  - The System bus connects the CPU and the main memory, i.e., (RAM).
- The bus provides a communication path for the data and control signals moving between the major components of the computer system.
- The system bus works by combining the functions of the three main buses:
  - System buses are of three types.

    A. Address bus

    B. Data bus

    C. Control bus.
- Each of the three buses has its separate characteristics and responsibilities.



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Address System Bus

- *An address bus transfers information about where the data should go.*
- *It carries memory addresses from the CPU to other components such as main memory and I/O devices.*
- The address bus is unidirectional.
- The address bus is used by the CPU or a direct memory access (DMA) enabled device to locate the physical address to communicate read/write commands.
- All address busses are read and written by the CPU or DMA in the form of bit.
- An address bus is measured by the amount of memory a system can retrieve.
- A system with a 32-bit address bus can address 4 gigabytes of memory space.
- Newer computers using a 64-bit address bus with a supporting operating system can address 16 exbibytes of memory locations, which is virtually unlimited.

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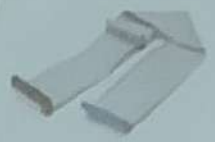
Data System Bus

- *A data bus is a system within a computer or device, consisting of a connector or set of wires, that provides transportation for data.*
- *A data bus is a computer subsystem that allows for the transferring of data from one component to another on a motherboard or system board, or between two computers.*
- It carries the data between the CPU and other components such as CPU, RAM, and I/O devices.
- Different kinds of data buses have evolved along with personal computers and other pieces of hardware.
- Each one is designed to handle so many bits of data at a time.
- *The amount of data a data bus can handle is called bandwidth.*
- A typical data bus is 32-bits wide. This means that up to 32 bits of data can travel through a data bus every second.
- Newer computers are making data buses that can handle 64-bit and even 96-bit data paths.
- At the same time they are making data buses to handle more bits, they are also making devices that can handle those higher bitrates.
- The data bus is bid



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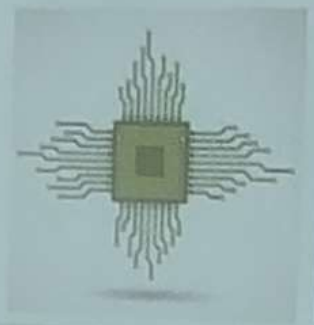
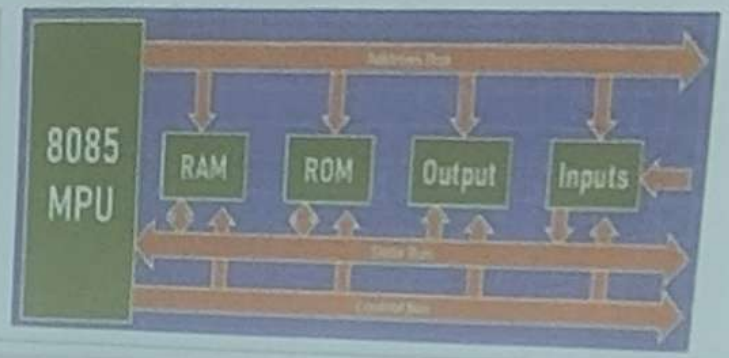
Control System Bus

- It carries control signals from the CPU to other components such as RAM and I/O devices.
- This bus is used to indicate whether the CPU is reading from memory or writing to memory.
- So control bus is a collection of signals that control how the processor communicates with rest of the system.
- The control bus also holds the clock's pulses.
- *It is also called the command bus.*
- The control bus is unidirectional.